TIPA Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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|---|---|---|---|---|
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## 2. Technology Overview

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| Title of Proposed Technology | High-Durability, High-Precision Transfer Stamp for Mass Transfer of Micro-LEDs<br>Micro-LED 대량 전사를 위한 고내구성·고정밀 전사용 스탬프 핵심 기술 개발 | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Electrics & Electronics* | | *Electrics & Electronics* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Display* | *Display manufacturing equipments* | *Semiconductor equipment* | *Other semiconductor equipments* |
| Concept of Proposed Technology | • This project aims to develop high-durability and high-precision transfer stamp technology applicable to Micro-LED mass transfer processes by combining Fraunhofer IAP's advanced polymer materials and surface/interface modification technologies with the ultra-precision mold design and processing technologies of one or more Korean industrial partners possessing precision machining and manufacturing capabilities.<br>• Among various Micro-LED chip assembly approaches, such as laser-based transfer, pick-and-place, and self-assembly methods, stamp-based mass transfer is considered the most suitable process for large-area and high-throughput Micro-LED display manufacturing. Stamp-based transfer enables simultaneous transfer of a large number of Micro-LED chips, offering clear advantages in productivity and scalability compared to serial transfer processes.<br>• In addition, stamp-based mass transfer reduces process complexity and equipment requirements compared to laser transfer, while avoiding the throughput limitations of pick-and- | | | |

place methods. By utilizing elastomeric materials with controllable adhesion, the process allows stable pick-up and release of ultra-small Micro-LED chips, supporting high transfer yield and accurate alignment, and is therefore regarded as a promising solution to current yield and throughput challenges in Micro-LED mass production.

- The transfer stamp is an elastomer-based hybrid mold/stamp system, designed to stably maintain strong yet precisely controlled adhesion during the pick-up and release of ultra-small Micro-LED chips, while simultaneously securing ultra-fine resolution and high alignment accuracy.
- Through polymer material innovation at Fraunhofer IAP, adhesive properties, elasticity, and thermal stability are improved,
- and by combining the precision manufacturing and system integration capabilities of the participating Korean institutions, the stamp is enabled to be repeatedly used over multiple transfer cycles while maintaining reliable performance.
- That is, the core concept of this technology is to fundamentally address the yield reduction and throughput limitations currently faced in Micro-LED manufacturing processes by realizing a reusable transfer stamp with significantly improved lifespan and adhesion reliability.
- Through improved material compositions and surface regeneration processes, the stamp can be applied to repetitive mass transfer processes without performance degradation and is integrated into a robust stamp design optimized for next-generation Micro-LED display mass production.

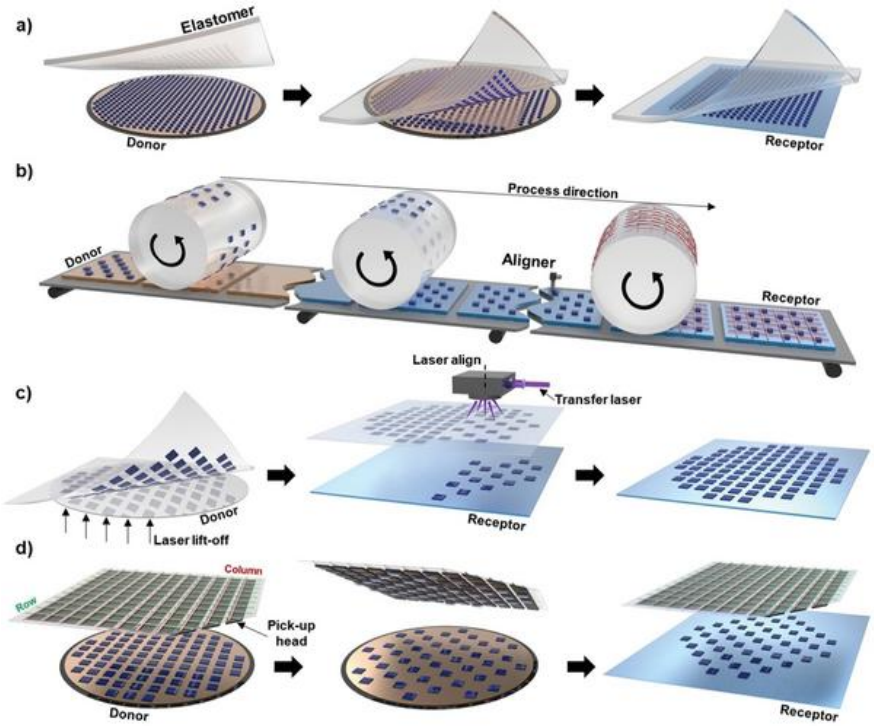


*Figure: Schematics of massive pick-up transfer methods. a) Elastomer stamp transfer method. b) Roll-to-roll or roll-to-plate transfer method. c) Laser-assisted transfer method. d) Electrostatic or electromagnetic pick-up transfer method. (ref (1))*

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| **Rationale for Joint R&D Project** | • Micro-LED mass transfer processes are widely recognized as the most critical technical bottleneck hindering the commercialization of Micro-LED displays. Various approaches, such as stamp transfer, laser transfer, and self-assembly methods, have been studied; among these, stamp-based mass transfer is evaluated as the most promising technology for mass production due to its capability for simultaneous large-area transfer and its relatively simple process structure.<br>• However, conventional transfer stamp technologies have limitations in that repeated use leads to issues such as degradation of polymer materials, deterioration of adhesion reliability, and surface contamination, resulting in transfer failures, yield reduction, and increased process costs. These issues are complex problems involving materials, processes, and design technologies, and are difficult to resolve by a single institution or a single technical field alone.<br>• Accordingly, this project essentially requires an international joint R&D structure in which Fraunhofer IAP, possessing world-class polymer materials and interface control technologies, and multiple Korean companies or a consortium with capabilities in precision machining, mold fabrication, and equipment integration, cooperate in a complementary manner.<br>• Fraunhofer IAP provides core fundamental technologies related to control of adhesion properties of transfer stamps, analysis of degradation mechanisms, improvement of material lifetime, and surface regeneration technologies, while the Korean lead institution implements these into manufacturable stamp structures and integrates and verifies them within processes and systems suitable for mass production environments.<br>• The technologies developed through this joint research are intended to be established as a technology base that is not dependent on a single company but can be disseminated across the domestic industry and further developed into an open technology platform that various domestic end-user companies can apply and commercialize in Micro-LED manufacturing processes.<br>• Through this, it is possible to reduce dependence on overseas sources for strategic core components, while simultaneously strengthening the technological self-reliance and competitiveness of the domestic display industry. |
| **Key Words** | 1. *Micro-LED*<br>2. *Mass Transfer*<br>3. *Transfer Stamp*<br>4. *Material Durability*<br>5. *Precision Mold* |

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 4) | End of the Project | Level (7 ) |
|---|---|---|---|---|
| R&D Objectives | **The R&D objective of this project is to secure core technologies for high-durability and high-resolution transfer stamps applicable to Micro-LED mass transfer processes. To this end, the following detailed technology development objectives are defined:**<br><br>1. Development of high-durability transfer stamp technologies that maintain stable adhesion and alignment performance even under repeated use<br>2. Securing polymer/elastomer material–based lifetime extension technologies that can suppress thermal and mechanical degradation of transfer stamps<br>3. Establishment of a reusable hybrid mold/stamp system combining precision mold structures with functional elastomers<br>4. Development of stamp surface regeneration processes capable of restoring adhesion properties in cases of contamination and performance degradation<br><br>**Through these developments, the ultimate goal is to improve Micro-LED transfer yield and to fundamentally resolve process cost increases caused by shortened replacement cycles of transfer stamps.** | | | |
| R&D Target Specifications | • **Resolution**<br>: Implementation of fine stamp structures compatible with ≤ 10 μm–class Micro-LED chips<br>• **Height variation of stamp patterns**<br>: ±5 μm<br>• **Shrinkage relative to stamp design (Total pitch)**<br>: ≤ 0.005%<br>• **Adhesion reliability of transfer stamps**<br>: Adhesion force variation within ±5% under repeated use<br>: Recovery of adhesion performance through cleaning and surface activation processes<br>: Maintenance of performance without damage to fine structures even after repeated regeneration processes<br>• **Reusable lifespan**<br>: Improvement of the number of reuse cycles by 30–40% compared to conventional transfer stamps in pick-and-place cycles | | | |
| R&D Plan | **Phase 1: 6 months**<br>Step 1: Material design and basic property evaluation (6 months)<br>• Design of high-durability elastomer and polymer material candidates<br>• Evaluation of key material properties, including adhesion, elastic modulus, and thermal stability<br>• Analysis of degradation mechanisms under repeated loading conditions<br>• Process design for reusability | | | |

- Establishment of a joint IP strategy

**Phase 2: 30 months**

Step 2: Transfer stamp structural design and sample fabrication (12 months)

- Design of ultra-precision mold structures optimized for Micro-LED arrays
- Fabrication of transfer stamp prototypes incorporating the materials
- Verification of initial transfer performance and alignment accuracy
- Development of process flows for reusability

Step 3: Lifetime extension and surface regeneration process development (12 months)

- Establishment of stamp contamination removal and surface activation processes
- Verification of repeatable regeneration capability and performance recovery level
- Evaluation of process compatibility and applicability of reusability processes

Step 4: Pilot demonstration and reliability evaluation (6 months)

- Process optimization for reusability
- Execution of repeated transfer tests in Micro-LED mass transfer environments
- Acquisition of yield, alignment accuracy, and lifespan data
- Demonstration of TRL 7–level prototypes and verification of commercialization feasibility

**Milestones**

- MS1 (Month 6)
  - Completion of high-durability elastomer/polymer material candidate selection
- MS2 (Month 18)
  - Completion of transfer stamp prototype fabrication and initial transfer performance verification for Micro-LEDs
- MS3 (Month 30)
  - Completion of lifetime extension technology verification for transfer stamps based on surface regeneration processes
- MS4 (Month 36)
  - Completion of TRL 7–level pilot demonstration in a Micro-LED mass transfer environment

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • Design and optimization of polymer/elastomer materials for transfer stamp repair<br>• Analysis of adhesion characteristics, thermal and mechanical stability, and degradation mechanisms<br>• Technical consultation and joint development of surface treatment and regeneration processes<br>• Support for scientific analysis and data interpretation of material and stamp performance |
| **Expected R&R of Korean SME Partner** | • Design of transfer stamp materials/structures and fabrication of ultra-precision molds<br>• Fabrication of material-applied stamp prototypes and process integration<br>• Application to mass transfer equipment and execution of pilot tests<br>• Evaluation of transfer yield, alignment accuracy, and reliability under repeated use<br>Linkage with domestic end-user companies and establishment of future commercialization strategies |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | • **Technical outcomes**<br>  ○ Securing core fundamental technologies for high-durability and high-reliability Micro-LED transfer stamps<br>  ○ Creation of new intellectual property related to materials, processes, and structural design<br>• **Commercialization and industrial outcomes**<br>  ○ Improvement of Micro-LED transfer yield and reduction of manufacturing costs<br>  ○ Establishment of a foundation for domestic production of transfer stamps and strengthening of the supply chain<br>  ○ Facilitation of entry of domestic SMEs into the Micro-LED industry and enhancement of their competitiveness<br>• **Economic and industrial ripple effects**<br>  ○ Acceleration of commercialization of next-generation displays (Micro-LED, AR/VR, automotive displays, etc.)<br>  ○ Reduction of dependence on overseas sources for strategic core process components<br>  ○ Strengthening of the global competitiveness of the domestic display industry<br>  ○ Promotion of joint intellectual property creation, standardization, and entry into the European market based on international collaboration |

## 6. References

(1) C. Linghu, S. Zhang, C. Wang, H. Luo, J. Song, Mass transfer for Micro-LED display: Transfer printing techniques, Semiconductors and Semimetals, Vol. 106, pp. 253–280, 2021.

(2) S. Zhang et al., Research Progress of Micro-LED Display Technology, Crystals, 13(7):1001, 2023.

(3) J. Zhao, Mass transfer strategies for MicroLED chip assembly, Int. J. Micro and Nano Manufacturing, 2025 (in press).

## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |